

YOURSTORY



snowflake

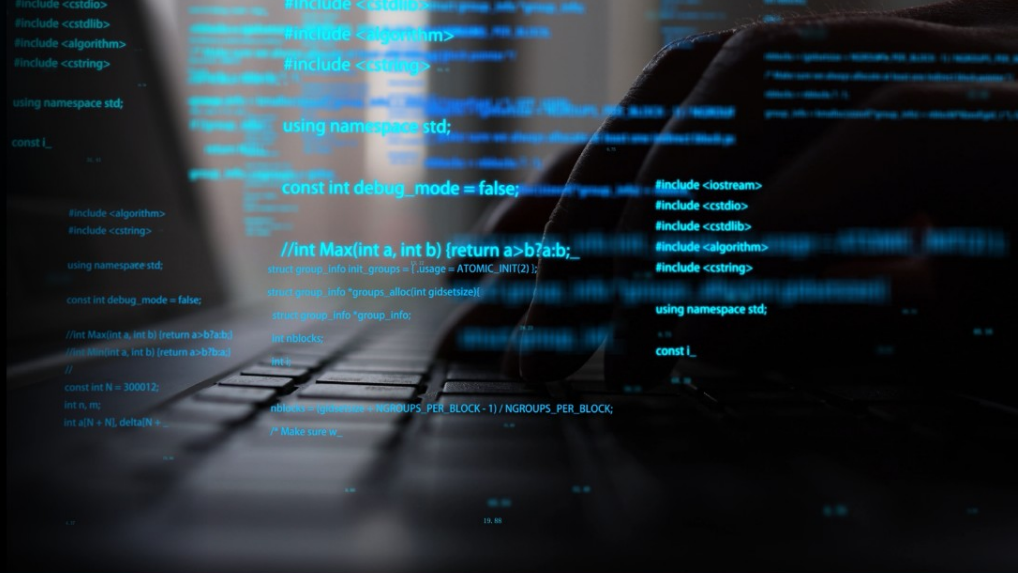
H2S



Coco CLI

HACKATHON

GCC Edition



Team Name : **MuleWatch**

Team Leader Name : **Gowtham Amarnath**

Team Size : **1**

Problem Statement : **Domain-Specific AI Copilot (BFSI - financial crime)**

Problem: scam money escapes through mule rings within hours



Business problem

Proceeds of cyber-fraud (investment, task-job and 'digital arrest' scams) are laundered through networks of mule accounts: collected, layered and cashed out at ATMs or crypto exchanges within hours.



Target user

Financial Crime Unit analysts and the Principal Officer's team at bank GCCs, who investigate alerts and file Suspicious Transaction Reports (STRs).



Pain today vs MuleWatch

Rule alerts are mostly noise (28-36% precision in our baseline) and every case is stitched together by hand. MuleWatch finds the whole ring, assembles the evidence and drafts a cited STR in minutes.



Industry context

Indian BFSI: real-time UPI / IMPS payments, KYC and PMLA reporting duties, and victim complaints via the 1930 helpline. Speed decides whether funds can be frozen before withdrawal.

MuleWatch: from suspicious alert to approved STR in minutes



01 Detect

Dynamic Tables compute 30-day behaviour. Rules, an ML model and a money + device graph find whole mule rings.



02 Investigate

The investigator agent builds an evidence pack from 8 sources: money trail, dwell time, devices, complaints, KYC, media.



03 Draft the STR

The compliance agent retrieves policy with Cortex Search, drafts the STR with clause citations and self-checks it.



04 Learn

The analyst approves or rejects. Each decision re-weights the rules and re-trains the model on a schedule.



Human-in-the-loop by design: the AI drafts, a named reviewer decides. Every AI action is written to an audit log, and personal data is masked for analysts.

Architecture: one Snowflake-native data flow

DATA SOURCES

Structured

Ledger, accounts, customers, victim complaints

Semi-structured (JSON)

Device telemetry, payment messages

Unstructured

KYC notes, news feed, policy & SOP documents

FEATURES

Dynamic Tables

TXN_ENRICHED
DEVICE_SHARING
CIRCULAR_FLOWS
PAYMENT_MSG_FLAT
ACCOUNT_FEATURES

Auto-refresh every 15 min

DETECTION (Snowpark)

Rules engine

RULE_CONFIG table

ML classifier

Model Registry

Graph rings

NetworkX components

CASES

Blended score

CORTEX AI AGENTS

Enrichment

AI_COMPLETE,
AI_CLASSIFY, AI_FILTER

Triage agent

Priority + SLA

Investigator agent

8-source evidence

Compliance agent

STR + Cortex Search

EXPERIENCE

Investigator app

Streamlit in Snowflake

Copilot

Cortex Agent + Analyst on a Semantic View

Automation

Tasks DAG, every 15 min

Learning loop analyst decision > FEEDBACK_LABELS > TUNE_RULES + model retrain (Tasks) **Governance** tag-based PII masking, AI audit log

How CoCo CLI skills and the modules plug together

COCO CLI PROJECT SKILLS (./cortex/skills)



\$mulewatch-deploy

Runs the SQL scripts in order, uploads the code, and fixes syntax drift between Snowflake releases.



\$mulewatch-investigate

Works a case from the terminal: evidence, STR quality checks, recommended decision for a human.



\$mulewatch-add-typology

Adds a new crime pattern across data, features, rules, policies and evaluation, tested offline first.

MODULES (each one swappable)

Module	Snowflake feature	Extend by
Ingestion	Tables + VARIANT	Pointing at core banking / complaint feeds
Features	Dynamic Tables	Adding a column = a new signal
Rules	RULE_CONFIG table	Tuning thresholds with no code
ML	Model Registry	Retraining from analyst feedback
Agents	Python procs + Cortex AI	Switching the LLM in APP.CONFIG
Copilot	Cortex Agent + Semantic View	Adding verified queries
App & ops	Streamlit + Tasks	Resuming / suspending the DAG

Impact: fewer, better alerts and investigations in minutes

67%

fewer alerts than rules alone

97.8%

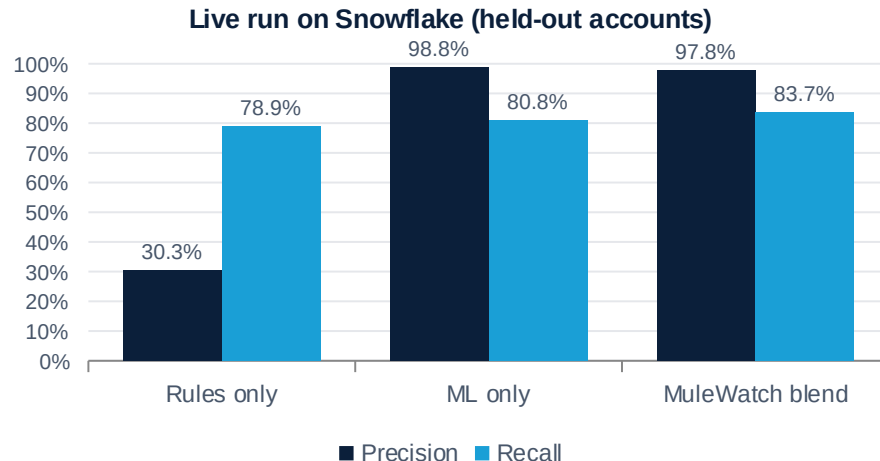
precision, vs 30.3% for rules only

83.7%

recall on held-out mule accounts (rules: 78.9%)

100%

of held-out rings found as one network



Minutes, not hours

First live run: 3 STRs drafted with 100% valid policy citations and 100% of figures traced to evidence.



Scales with Snowflake

~0.9M transactions scored on an X-Small warehouse. AI runs only on alerted accounts to control cost.



Beyond the demo

New typologies via a CoCo skill; the same pattern fits insurance and trade fraud; live feeds via Snowpipe.

See it live: demo flow and links

1

Command Center: live KPIs and precision versus the rules baseline

2

Inject a live mule-ring attack, then run the detection pipeline

3

Investigation: ring network graph, money timeline and AI findings

4

Draft the STR with policy citations, edit it, and approve

5

Ask MuleWatch: the copilot answers on cases, rings and policy

LINKS



Public GitHub repository

github.com/Gowtham619/mulewatch



Deployed app (Streamlit in Snowflake)

[Open MULEWATCH_APP in Snowsight](#)



Snowflake features used

Dynamic Tables, Snowpark, Model Registry, Cortex AI functions, Cortex Search, Semantic Views, Cortex Agents, Tasks, Streamlit, Horizon masking

YOURSTORY



1125

Coco CLI

HACKATHON

GCC Edition

THANK YOU



```
#include <algorithm>
#include <string>

using namespace std;

const int debug_mode = false;

#include <algorithm>
#include <string>

using namespace std;

const int debug_mode = false;

//int Max(int a, int b) {return a>b?a:b;}
//int Min(int a, int b) {return a<b?a:b;}
//
const int N = 300012;
int n, m;
int a[N + N], delta[N + N];

// Make sure w
```